
Topic 9

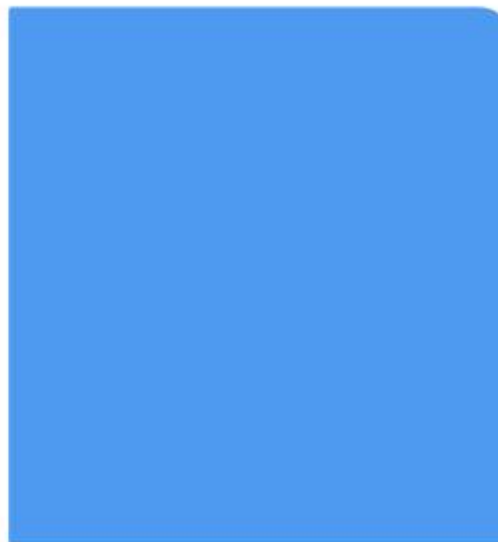
Scientific reasoning



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Science

27



Confirms

14



Disconfirms



Overview

- Introduction
- Fixation of belief
- What is science?
- The logic behind science
- Scientific reasoning
- Falsifiability criterion
- Two theories

Introduction



Introduction

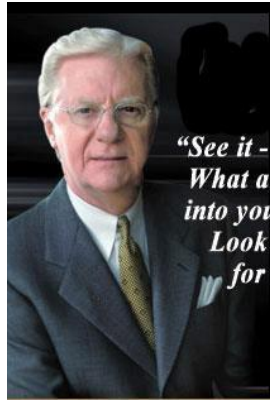
- *the Secret*
 - The *law of attraction*: “like attracts like.”
 - The *law of attraction* is presented as a law of nature that is science based.
 - If you don't believe in it (consciously or unconsciously), then the universe realizes that it does not work for you.
 - Furthermore... a lot of money is earned...

Introduction

**Double Your Income,
Not Your Workload.**

Free Workshop
LONDON
7th June 2018

Free Admission . Limited Spaces . Register Now



*"See it - Believe it - Do it!
What are you attracting
into your life right now?
Look at your results
for the answer."*

JOHN ADAMOFF • FEARLESS INVESTMENT

**TRAIN
YOUR
BRAIN**
TO MAKE MORE
MONEY!



THE
SCIENCE of GETTING RICH
MICHAEL BECKWITH • BOB PROCTOR • JACK CANFIELD

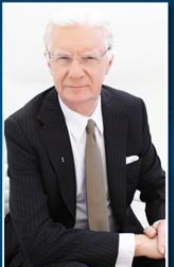


**ATTRACT
THE PERSON
YOU DREAM ABOUT
NOW!**

**CHANGING YOUR
SELF-IMAGE,
LEADERSHIP, AND THE
ABC'S OF SUCCESS**

- With Guest Bob Proctor

THE MODEL SHOW
Episode 111



Introduction



Your Online Resource for Law Of Attraction Certification!

Testimonials

"Dr. Rubino, I wish I had gotten this training much earlier in my life. It would have saved me 40 years of struggle. Thanks for opening my eyes!"

-Peter M.

"Your material sounded so good that I doubted you could deliver what you promised. It has and more. This course is amazing. The down to earth principles you share are life transforming. I thought that I had plenty of self-esteem, but you showed me how I was living a lie in so many areas of my life. Thanks to this course, I have found my authentic self."

Maggie C.

LAW OF ATTRACTION BASIC PRACTITIONER C

Location: Internet

Dates: You may start anytime.

Pace: Three weeks is the shortest allowable completion time. Eighteen months is the longest allowable completion time.

Cost: \$675 (includes 5 training manuals)

Cost: ~~\$475~~ \$275 (includes 5 training manuals)

Registration deadline: None. Register anytime.

Cancellation policy: Cancellations are not allowed.

Instructors: Dr. Joe Vitale, Star Of The Hot Movie "The Secret" Author Of Over 50 Books Including The Attractor Factor And The Key & Dr. Steve G. Jones, Ed.D., NLP Master Trainer, Board of Directors Member: American Lung Association (LA Chapter), Member: American Board of Hypnotherapy, Member: National Guild of Hypnotists
• Founder: American Alliance of Hypnotists

Text Books (E-Book format): As part of the course (included in the total price of this course), you will receive 5 study manuals (each approximately 20 pages in length) by Dr. Joe Vitale and Dr. Steve G. Jones, Ed.D., M.Ed, NLP Master Trainer:

FAQs



Introduction

- The ideas outlined in *the Secret*
 - sound plausible
 - appear to be scientific
- But there is no scientific justification to believe in these ideas.
 - Why not?

Hindsight Bias and Overconfidence

- **Hindsight Bias**

- the “I-knew-it-all-along” phenomenon.
- After learning the outcome of an event, many people believe they could have predicted that very outcome.
- Anything seems common-place once explained.

“Life is lived forwards, but understood backwards.”

Philosopher Søren Kierkegaard, 1813–1855

- **Overconfidence**

- Sometimes we think we know more than we actually know.
- Both hindsight bias and overconfidence lead us to overestimate our intuition

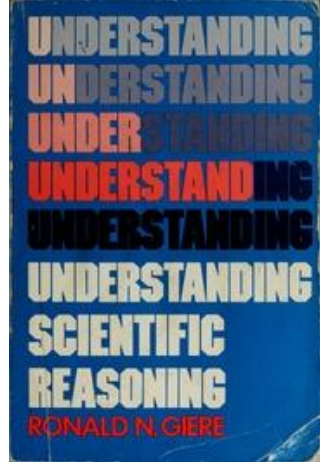
Perceiving order at random things



“...with a large enough sample, any outrageous thing is likely to happen,”

statisticians Persi Diaconis and Frederick Mosteller (1989)

Fixation of belief



- Belief
 - An attitude of some person toward a particular claim
 - For example
 - *I believe that "like attracts like as stated in the law of attraction".*
 - *I believe that "intelligent aliens visit earth on a regular basis".*
 - *I believe that "the current rates of climate change is caused by humans emitting greenhouse gases".*
 - *I believe that people are inherently good/bad.*
 - *I believe that "time dilation* exists".*

**time dilation = a difference in the elapsed time measured by two clocks, e.g., due to them having a velocity relative to each other (Albert Einstein)*

Fixation of belief

- Belief
 - An attitude of some person toward a particular claim
- Claim (or statement)
 - true
 - false
- There is no direct connection between belief and truth
 - believing that a given claim is true does not provide information about the truth or falsity of the claim itself.

Fixation of belief

- For example,
 - *Like attracts like as stated in the law of attraction.*
 - *Intelligent aliens visit earth on a regular basis.*
 - *The current climate change is caused by humans emitting greenhouse gases. **
 - *Time dilation exists. **
 - *The world is designed by an Intelligent designer (intelligent design theory).*
 - *Life on earth evolved by natural selection (evolution theory). **

**These claims are scientifically justified*

Fixation of belief

- Why is the claim "time dilation exists" (Einstein) scientifically justified?
- Why is the claim "intelligent aliens visit earth on a regular basis" not scientifically justified?

Fixation of belief



It is difficult to imagine that clocks run slow when moving

Fixation of belief



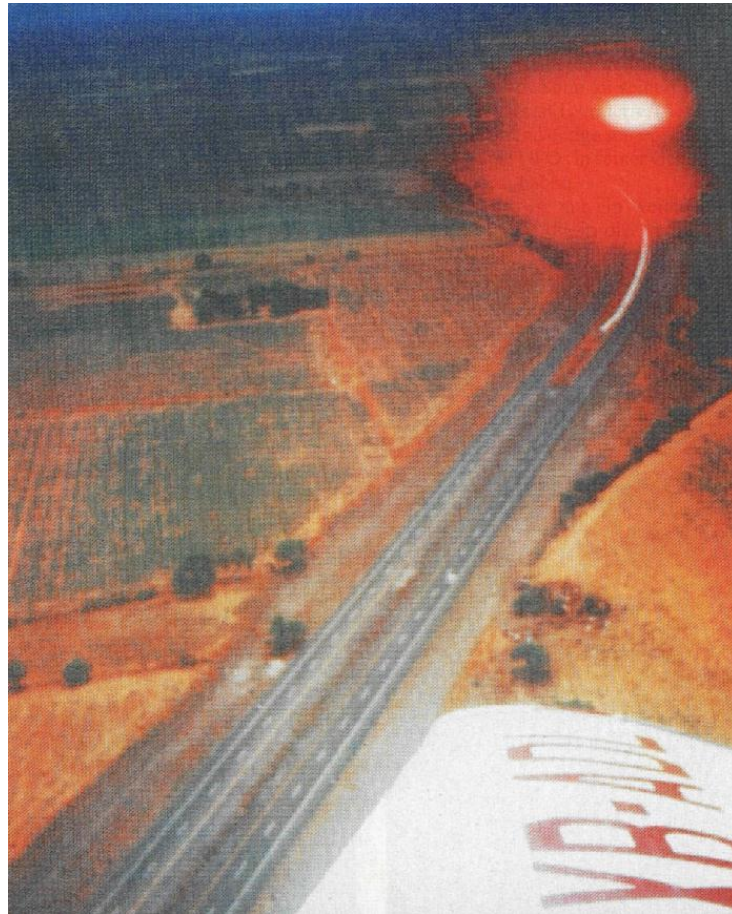
There are many reports of encounters with aliens (on the internet)

Fixation of belief



There are many pictures of “UFOs”

Fixation of belief



There are many pictures of “UFOs”

Fixation of belief

- Why is the claim "time dilation exists" (Einstein) scientifically justified?
 - The claim is based on the proper use of the scientific method
- Why is the claim "intelligent aliens visit earth on a regular basis" not scientifically justified?
 - The claim is not based on the proper use of the scientific method

Fixation of belief

Non-scientific methods

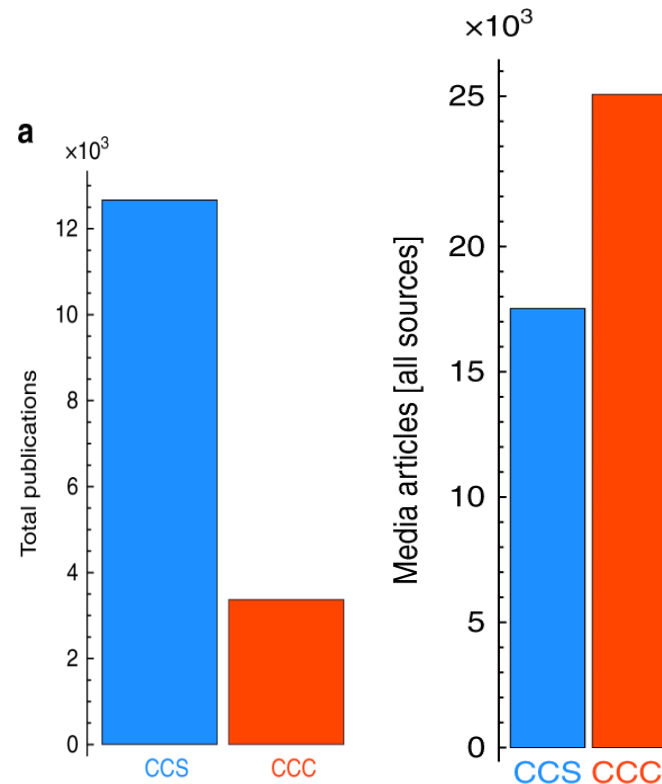
- What are the methods to fix beliefs?
 - Personal experience
 - e.g., "Global warming isn't real because I was cold today!" (Stephen Colbert joking on Twitter)
 - Intuition
 - "President Donald Trump told the Associated Press that he has a "natural instinct for science" that informs his understanding of climate change [which according to him is not human-made]." (<https://www.politico.com/story/2018/10/17/trump-instinct-climate-change-910004>)
 - Authority
 - Only a fortnight after the UN Intergovernmental Panel on Climate Change (IPCC) issued what is arguably the most comprehensively damning report on the fate of the planet in the history of science, Trump assured people they don't have to worry [about climate change]." (<https://www.sciencealert.com/trump-claims-natural-instinct-for-science-latest-rebuttal-climate-change-global-warming-ipcc>)

Fixation of belief

- Consensus among scientists: Climate change is caused by humans
 - So, what's the evidence?
 - Basic physics that shows us that CO₂ absorbs heat (research ~ 1820s)
 - Ruling out natural factors that can influence climate like the sun and ocean cycles (research ~1830s)
 - Simple chemistry – when we burn carbon-based materials, carbon dioxide (CO₂) is emitted (research ~1900s)
 - Monitoring climate conditions to find that recent warming of the Earth is correlated to and follows rising CO₂ emissions (research ~ 1930s)
 - Measuring CO₂ in the atmosphere and trapped in ice to find that it's increasing, with levels higher than anything we've seen in hundreds of thousands of years (measurements ~1950s)
 - Chemical analysis of the atmospheric CO₂ that reveals the increase is coming from burning fossil fuels (research ~1950s)
 - Employing computer models to run experiments of natural versus human-influenced simulations of Earth (research ~1960s).
 - Basic accounting of what we burn, and therefore how much CO₂ we emit (data collection ~ 1970s). etc. etc.

Fixation of belief

- Consensus among scientists:
Climate change is exacerbated by human activity



CCS = Climate Change **Scientists**

CCC = Climate Change **Contrarians**

Fixation of belief

- The scientific method fixes belief on the basis of *empirical research*
 - It reaches conclusions by combining:
 - proper reasoning (i.e., use of logic arguments)
 - systematic empiricism (i.e., use of verifiable evidence based on experience with the world)

Can we study **EVERYTHING** scientifically?

- Is there life after death?
- How many angels can dance on the head of a pin?

Solvable Problems

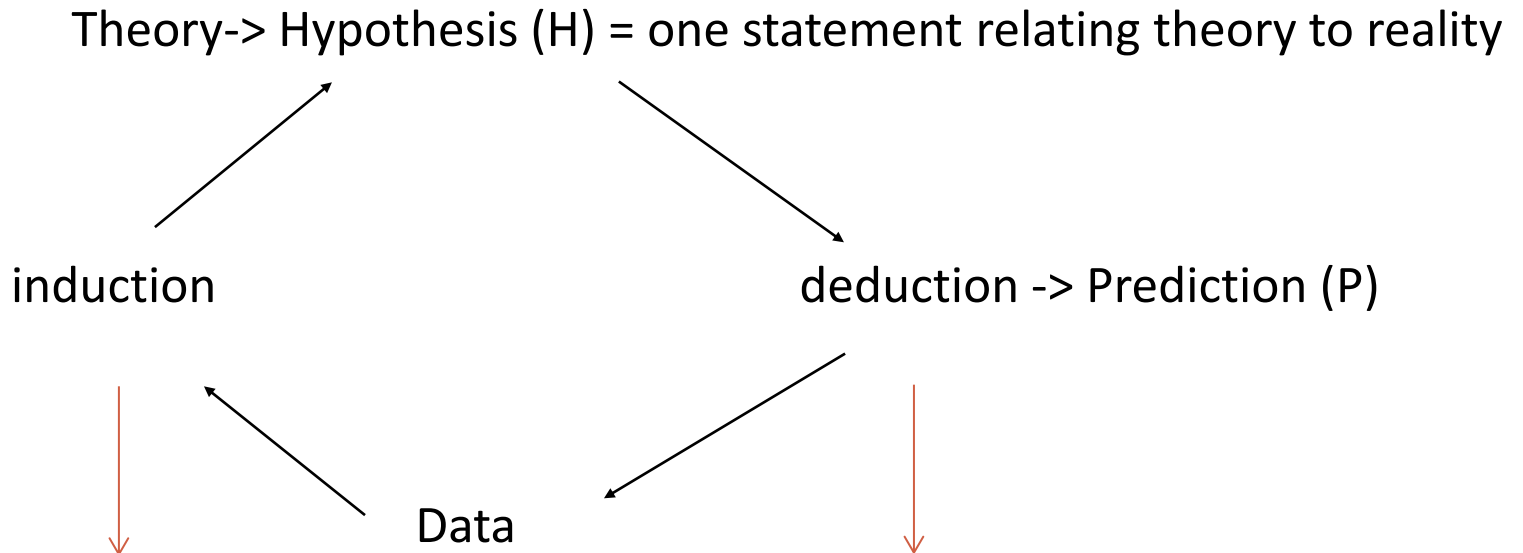
- Science only deals with solvable problems.
- Researchers can investigate only those questions that are answerable, given current knowledge and research techniques.
- In sum, to be considered scientific:
 1. the questions addressed must be potentially solvable
 2. observations must be systematic and empirical
 3. research must be conducted in a manner that is publicly verifiable
- **Pseudoscience** involves evidence that masquerades as science but that fails to meet one or more of the three criteria used to define science.

How Do Psychologists Ask & Answer Questions?

- Psychologists, like all scientists, **use the scientific method to construct theories** that organize, summarize and simplify observations.

What is science?

- Empirical cycle (theory-data cycle)



From specific (data) to general (theory/hypothesis)

Jan, Marie, and Willy are older than 17

Jan, Marie, and Willy are first-year students

Thus: All first-year students are older than 17 (?)

From general (theory/hypothesis) to specific (data)

All first-year students are older than 17

Jan is a first-year student

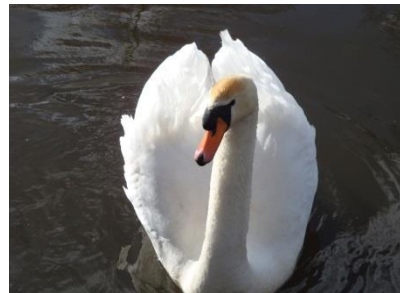
Thus: Jan is older than 17

What is science?

- **Induction**
 - Specifying a general rule based on specific observations
 - the conclusion is often plausible but not inescapable

What is science?

- the first swan in the park is white
 - the second swan in the park is white
 - the third swan in the park is white
 -
 - the last observed swan in the park is white
-
- SO ... all swans in the park are white



What is science?

- All swans in the park are white
- A swan is swimming in the park pond
- SO ... this swan is white



What is science?

- **Deduction**

- Specifying a conclusion about something specific on the basis of a general rule
- the conclusion follows inevitably from the assumptions / premises

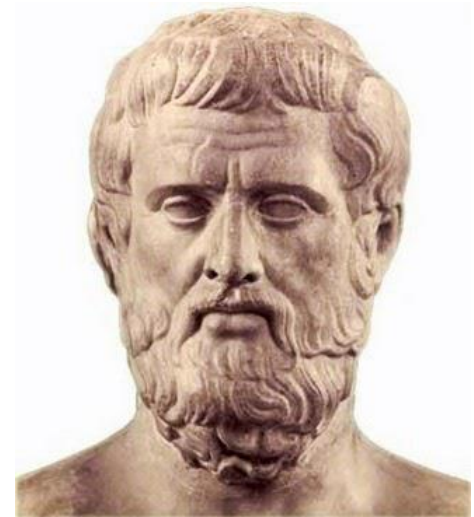
Inductive versus deductive

- **Deductive reasoning:**
 - top-down, using things you already know (rationalism)
 - Infinity exists, given the fact that you can always add a number to the previous number
- **Inductive reasoning:**
 - bottom-up, using observations (empiricism)
 - Infinity cannot (never) be proven because *nobody can count to infinity*

Deductive reasoning

Aristotle (384 – 322 BC)

- **Syllogism:**
 - 2 premises → conclusion
 - Valid or invalid
 - does the conclusion hold, given the premises?
 - is not the same as true or false!
- Two types of syllogisms:
 - Categorical syllogism (All A are B....)
 - Conditional syllogism (If A then B...)



What is science?

- Scientific method
 - A method to provide good justification for the belief that a claim or statement (hypothesis) is true or false.
- Goal of science
 - To find good justification for the belief that a claim or statement (hypothesis) is true or false.
 - by means of a **conditional argument**

The logic behind science

- Conditional argument

A = antecedent

B = consequent

Affirming the antecedent

If A then B

A

Thus B (valid)

Modus ponens

} Premises

If it rains, I stay at home

It rains

} Conclusion

I stay at home

Denying the consequent

If A then B

not B

Thus, not A (valid)

Modus tollens

} Premises

If it rains I stay at home

I do *not* stay at home

} Conclusion

It is not raining

Deductive reasoning

Conditional syllogism

- If P then Q
Not P
Therefore *not* Q

Framing is important!

If it rains I stay at home
It is not raining

I do not stay at home

If I live in Amsterdam, I live in the NL
I do not live in Amsterdam

I do not live in the Netherlands

Denying the antecedent

Beware, not a valid argument!

If you are a proofreader, you have a job.

You are not a proofreader, you do not have a job.

DEDUCTIVE REASONING

CONDITIONAL SYLLOGISM



- If P then Q
Q
Therefore P

Framing is important!

If it rains I stay at home
I stay at home

It rains

If I live in Amsterdam, I live in the NL
I live in the NL

I live in Amsterdam

Affirming the consequent

Beware, not a valid argument!

illicit conversion: If p then q $\rightarrow$ If q then p

40% of the criminals is an immigrant $\rightarrow$ 40% of the immigrants is criminal

The logic behind science

- Conditional argument

Affirming the consequent

If A then B

B

Thus A (invalid)

} Premises

} Conclusion

If it rains, I stay at home
It is not raining

I do not stay at home

Denying the antecedent

If A then B

not A

Thus, not B (invalid)

} Premises

} Conclusion

If it rains, I stay at home
I stay at home

It rains

The logic behind science

- Conditional argument

Affirming the antecedent

If A then B

A

Thus B

valid

Denying the consequent

If A then B

not B

Thus not A

Denying the antecedent

If A then B

not A

Thus not B

invalid

Affirming the consequent

If A then B

B

Thus A

The logic behind science

- The type of argument used to justify or falsify a scientific hypothesis is the **same across different scientific disciplines**.
 - The conclusion of the argument will be either the hypothesis or its negation.
- *Denying the consequent*
If A then B
not B

Thus, not A (valid)

Hypothesis

A **hypothesis** is a **testable prediction**, often prompted by a theory, to enable us to accept, reject or revise the theory.

For example:

One hypothesis from the Social Learning Theory:

People who watch violent films display more aggressive behavior.

Research

Research would require us to administer surveys or experiments.

For example:

Conduct a research study to examine if there is a relationship between frequency of watching violent films and aggressive behaviors.

The Scientific Method

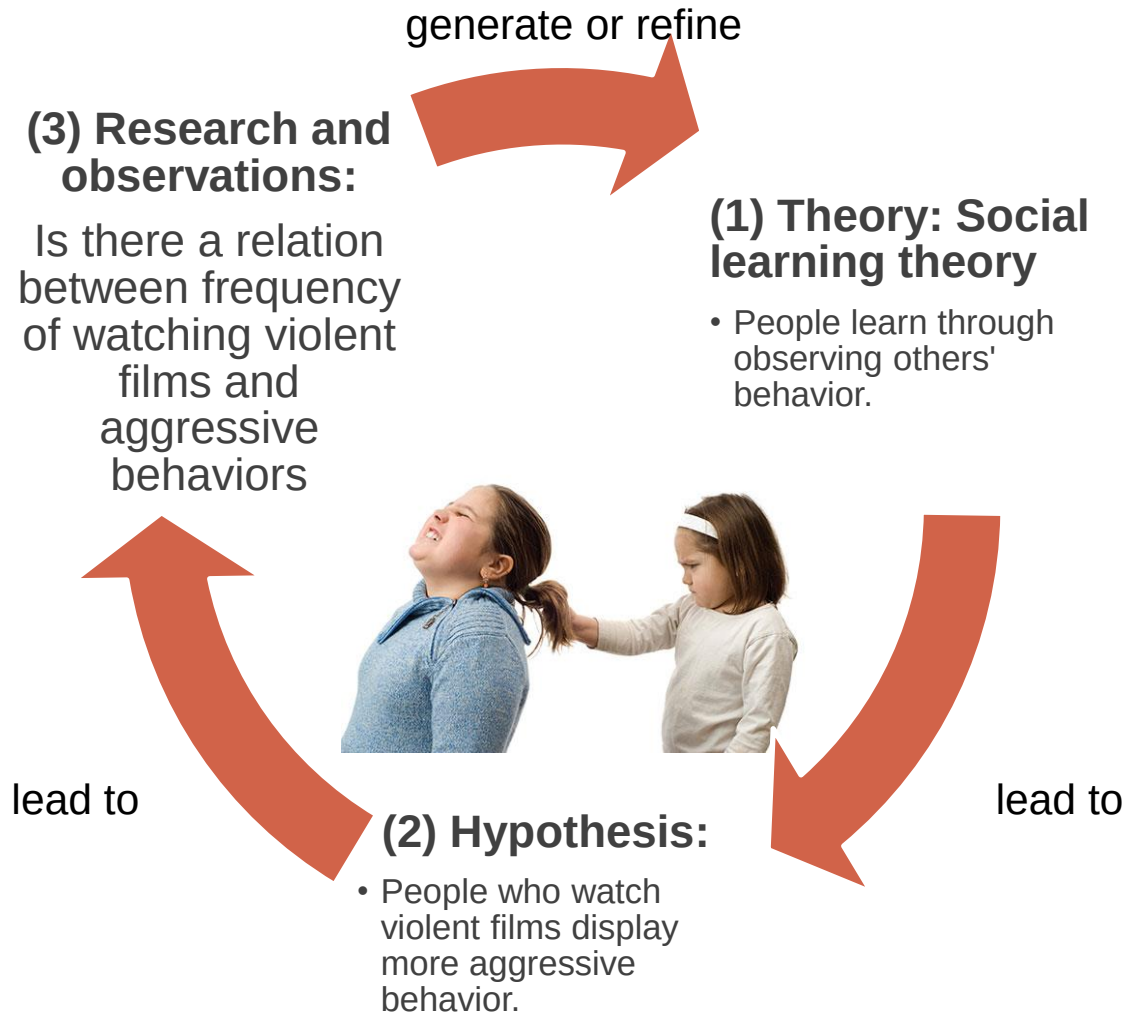
- **Operational Definition**

- a statement of procedures (operations) used to define research variables
 - How you are going to measure aggression is your operational definition for aggression.

Example

- Violent movies and aggression
- Hypothesis?
 - There is a relationship between frequency of watching violent films and aggressive behaviors
- Operational definition of variables?
 - **Exposure to violent movies:** number of times a person watches violent movies in a week in the last 3 months
 - **Aggression:** Frequency of behavior that aims to physically harm other people.

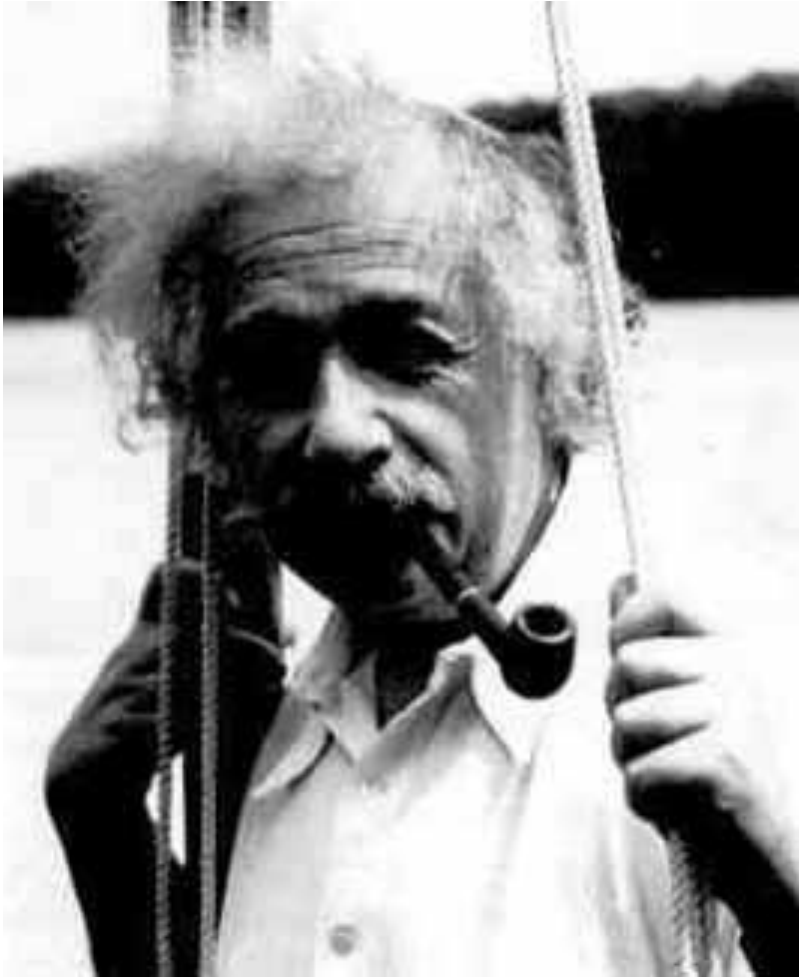
Research Process



The Scientific Method

- **Replication**
 - Repeating the essence of a research study to see whether the basic finding extends to other participants and circumstances
 - Usually with different participants in different situations

Scientific reasoning



Scientific reasoning

- Theory
 - *Einstein's theory of relativity*
- Hypothesis (H)
 - *Time dilation exists*
- How to test this hypothesis?
 - Is the hypothesis true?
 - Is the hypothesis false?

Scientific reasoning

- Experiment
 - *Experiment with atomic clocks in airplanes and on earth.*
 - *Assumption: time registration is reliable.*
- Specification of prediction
 - *Clocks in airplanes run more slowly than the clocks on earth.*

Scientific reasoning

- Hypothesis (H)
 - *Time dilation exists.*
 - Initial conditions (IC)
 - *Experiment with atomic clocks in airplanes and on earth.*
 - Auxiliary assumptions (AA)
 - *Registration is reliable.*
-
- deduction*
- Prediction (P)
 - *Clocks in airplanes run more slowly than the clocks on earth.*

If [H and IC and AA] then P

Scientific reasoning

- Prediction (P) turned out to be true!!
 - *Clocks in airplanes run more slowly than the clocks on earth.*

If [H and IC and AA] then P
P

Thus [H and IC and AA]

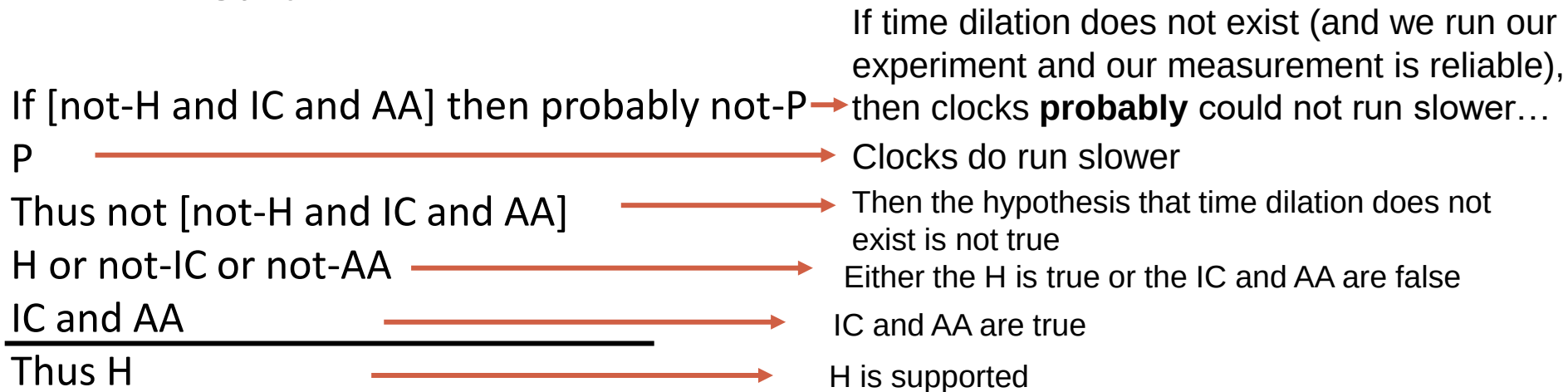
If time dilation exists, clocks should run slower...
Clocks run slower

Then time dilation exists

But this is an invalid argument
(affirming the consequent)!!

Scientific reasoning

- Prediction (P) turned out to be true!!
 - *Clocks in airplanes run more slowly than the clocks on earth.*



valid argument (inductively)

Scientific reasoning

- 2 conditions for a good test of the hypothesis

1 - If [H and IC and AA] then P

2 - If [not-H and IC and AA] then probably not-P

H	=	Hypothesis
IC	=	Initial Conditions (<i>e.g., description of state at beginning of experiment</i>)
AA	=	Auxiliary Assumptions (<i>e.g., registration is reliable</i>)
P	=	Prediction (follows deductively from H and IC and AA)

Scientific reasoning

- Justifying a theoretical hypothesis (inductively)

If [not-H and IC and AA] then **probably** not-P

P

Thus: not [not-H and IC and AA]

H or not-IC or not-AA

IC and AA

Thus: H

Probably

Scientific reasoning

- Refuting a theoretical hypothesis (deductively)

If [H and IC and AA] then P

not-P

Thus: not [H and IC and AA]

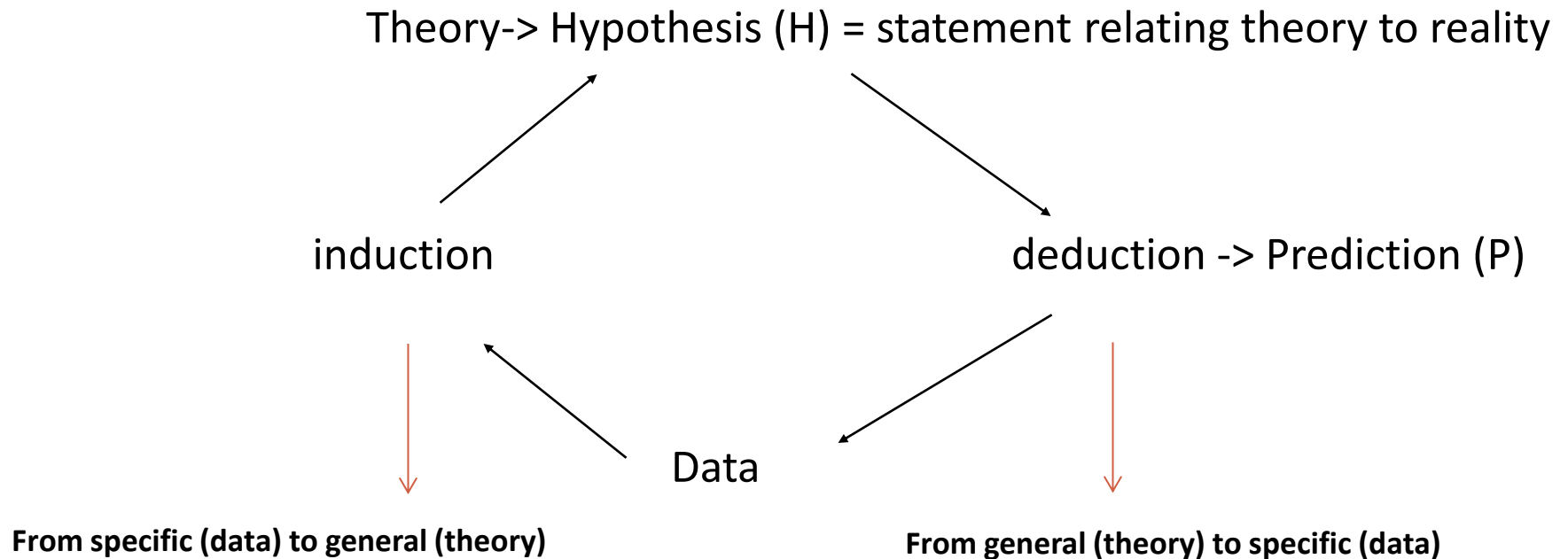
not-H or not-IC or not-AA

IC and AA

Thus: not-H

What is science?

- Empirical cycle (theory-data cycle)



Scientific reasoning

- Hypothesis (H)
 - *Time dilation exists.*
 - Initial conditions (IC)
 - *Experiment with atomic clocks in airplanes and on earth.*
 - Auxiliary assumptions (AA)
 - *Registration is reliable.*

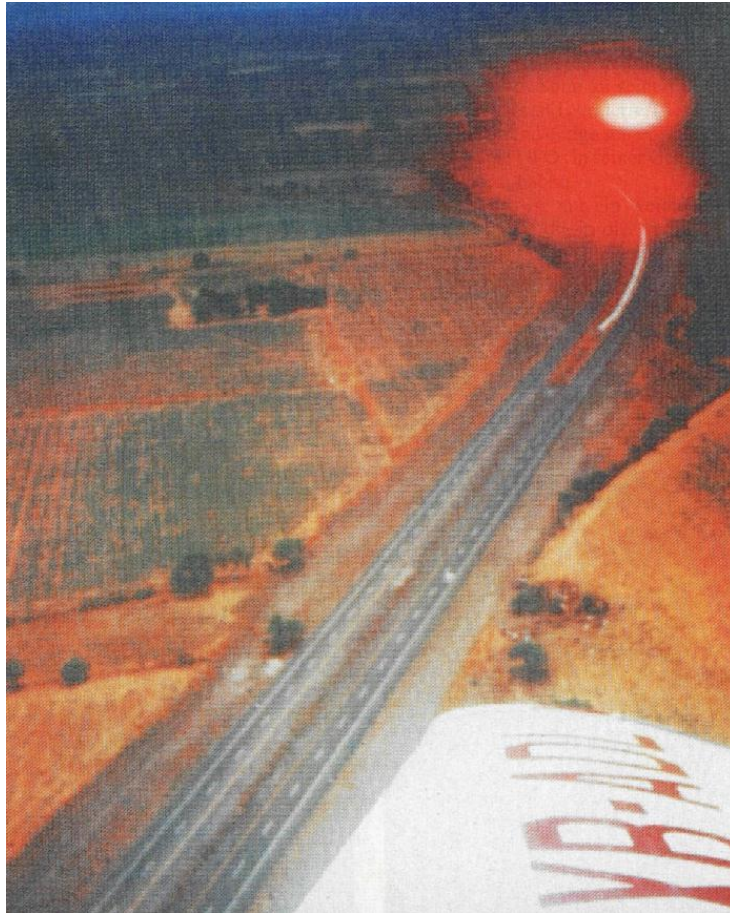
 - Prediction (P)
 - *Clocks in airplanes run more slowly than the clocks on earth.*
1. If [H and IC and AA] then P
 2. If [not-H and IC and AA] then probably not-P

Scientific reasoning

- Hypothesis (H)
 - *Intelligent aliens visit earth on a regular basis.*
 - Initial conditions (IC)
 - *Current situation on earth.*
 - Auxiliary Assumptions (AA)
 - *Registration is reliable and the aliens travel by visible UFOs.*

 - Prediction (P)
 - *Now and then somebody sees something similar to an UFO.*
1. If [H and IC and AA] then P
 2. If [not-H and IC and AA] then probably not-P

Scientific reasoning



This picture can also be the result of something else than H

Scientific reasoning

- Hypothesis (H)
 - *The law of attraction (the secret).*
- Initial conditions (IC)
 - *Someone thinks positively and wishes to become a millionaire.*
- Auxiliary Assumptions (AA)
 - *The person truly believes in the law of attraction.*

- Prediction (P)
 - *The person will become a millionaire.*

1. If [H and IC and AA] then P

2. If [not-H and IC and AA] then probably not-P

Scientific reasoning

- Result
 - not P

If [H and IC and AA] then P
not-P

Thus: not [H and IC and AA]
not-H or not-IC or not-AA
IC and H

Thus: not-AA

This is valid but not scientifically
justified!!

**because the hypothesis HAS TO BE
the conclusion**

Falsifiability criterion

Science



- People believe statements which
 - can be demonstrated to be false.
 - can be demonstrated to be ***most likely*** true.
 - cannot be demonstrated to be false or true.

Falsifiability criterion

- To find good justification for the belief that a hypothesis is true, the hypothesis needs to be potentially falsifiable.
 - has to be precise too
- Scientific hypotheses need to generate predictions that might not be true –*the falsifiability criterion* (Popper, 1961)

Falsifiability criterion

- If we test our ideas, we can know when we are wrong. And if we know when we are wrong, we can correct our mistakes. If we don't, we can't.
- *“Those among us who are unwilling to expose their ideas to the hazard of refutation do not take part in the game of science.” (Popper, 1934)*
- *“The essence of science is making mistakes in public.” (Dennet, 1995)*

Two theories

- Life on earth evolved gradually by natural selection (Darwin) - Evolution theory
- Life on earth is created by an intelligent designer - Intelligent Design Theory

Two theories

- Evolution theory (Darwin, 1859)
 - *“Life on earth evolved gradually beginning with one primitive species -perhaps a self-replicating molecule – that lived more than 3.5 billion years ago; it then branched out over time, throwing off many new and diverse species; and the mechanism for most (but not all) of evolutionary change is natural selection.”* (Jerry Coyne, 2009)

Two theories

- Evolution theory (Darwin, 1859)
 - Some predictions
 - The earth should be at least hundreds of million years old.
 - Precambrian fossils
 - *Darwin wrote in 1859 that the total absence of fossils in Precambrian rock was "inexplicable" and that the lack might "be truly urged as a valid argument" against his theory. When such fossils were found, starting in 1953, it turned out that they had been abundant all along. They were just so small that it took a microscope to see them.*
 - etc. etc.

Two theories

- *Intelligent Design* (e.g., William Dembski, 1999)
 - There are facets in life such as biochemical processes in living cells and interactions between genes that are so complex that it is unimaginable that these systems arose as a result of natural selection only. Evolution theory does not provide a satisfying explanation for crucial events (such as the beginning of life and the emergence of new species). These events occurred because an intelligent power has intervened and steered the processes into the right direction.

Two theories

- *Intelligent Design* (e.g., William Dembski, 1999)
 - Some predictions
 - ????

Two theories

- The evolution theory of Darwin is a scientific theory because it generates specific predictions rendering the theory potentially falsifiable.
- The Intelligent Design theory is not a scientific theory because it does not satisfy the falsifiability criterion.

Two theories

- However, everybody is free to believe in Intelligent design. It is possibly true.
 - Believing in intelligent design is a matter of faith and has nothing to do with science
 - Intelligent Design theory has not been repeatedly submitted to scientific testing.
- An established theory in science is based on far more than faith
 - A theory in a scientific sense is a conceptual structure that is supported by a large and varied set of data.



Graphs can lie to you

Addendum

“In an age where information can be shared in less than a minute, false information spreads fast. Times of fear and uncertainty provide a fertile ground for disinformation to grow. In other words, as the Covid-19 virus spreads around the world, so does another disease: fake news... We can check information over and over again, but we should also listen to sound advice and facts, no matter if they reinforce what we believe or not. As Ursula von der Leyen says in her address to Europe’s citizens: “Those spreading disinformation harm you. Disinformation can cost lives. But together, we can set the record straight”.



KEEP

CALM

this is the

END OF MY

LECTURE